Based on GAma Modeling Language

Internship "research and development"

A report by

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Contents

Acknowledgments **4**

Introduction **5**

1 Psychological aspect **6**
1.1 Psychology of human behavior 6
1.2 Human behavior in emergency situation 7
1.3 Models governing agents in crisis situations 8

2 Simulation of decision making **11**
2.1 Graphical interface of the simulation 11
2.2 Variables used in the simulation 12
2.3 Other files used 12
2.4 Vision coding 13

Conclusion **15**

References **16**

Abstract **17**

List of Figures

1.1 Order of preference for the selection of exits . . . 7
1.2 Number of evacuated people during the time . . . 8
1.3 Bayesian belief network . . . 9
1.4 Microscopic decision-making models . . . 9
1.5 COEX Mall . . . 10

2.1 Graphical interface of the simulation . . . 11
2.2 Shapefiles . . . 12
2.3 Horizontal and Vertical cases . . . 13
2.4 Diagonal case . . . 14
2.5 Random cases . . . 14

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Introduction

Decision-making is a major factor in understanding human behavior during an emergency. According to the different articles in this review, behavior is a complex phenomenon that interacts with different environmental factors.

Indeed, there is a real psychological study in terms of decision-making that will involve complex actions, due to stress for example. But there are also models that emerge to simulate these situations, especially in the current context, which are capable of modeling individual and crowd human behaviors.

For the purpose of prevention and safety of its users and collaborators, a large retail company wishes to engage in the implementation of scientific research aimed at better understanding the psychological aspects of human behavior in risk situations, and especially during evacuation situations.

The objective of this R & D mission is to study the possibility of modeling and simulating the perception of the functionality of situations.

From a perspective of situational functionality, I will rely on the embodied and situated approach of decision-making and the underlying psychological theories.

I propose to use the GAMA platform as a computer support for modeling and simulation. GAMA is a development environment for creating agent-based simulations: GAMA website

Chapter 1

Psychological aspect

1.1 Psychology of human behavior

The cognitive abilities of an agent considered in isolation, that is to say out of all interactions with his environment (located action) or with other agents (distributed cognition), are weaker than if he could support these elements to increase cognitive abilities. In fact, an agent whose abilities are limited (limited rationality) can achieve his goal in an environment because he has the cognitive resources of this environment and that he solves certain problems in interaction with other agents. In addition, it is said that the rationality of an agent is limited when his cognitive abilities prevent him from having optimal behavior. [1]

Depending on the situation and the environment, some of our cognitive abilities work differently:

- Perception: The principle of affordance is emerging because I will not perceive the properties of objects around us but rather the possibilities of actions that offer us these objects. Moreover, we can perceive our environment differently according to our interactions with it, for example when we can reach an object using a laser, we perceive the distance shorter than when we do not have a laser. [2]

- Action: There is a difference between planning and reaction, indeed, even if our actions are planned there will always be reactions to the environment around us. For example, before going down a ski slope, an agent plans his race but once in action he has to react to the various constraints with the help of his know-how. Moreover, the repetition of gestures and the habit influences our way of perceiving the world and therefore our reactions are unconsciously influenced.

- Language: To interpret an expression, you must always pay attention to the context of utterance, for example.

- Memory: We also use the environment to remind us of a task. For example, when we have an appointment, we note it in a diary or in our phone. Our knowledge uses all our senses that simulate an action and then add it to our memory. Moreover, we do not memorize the past actions but rather simulate the potential future actions, it is the "I act, so I think". [3]

The situated cognition is also interesting for the human behavior, indeed the group of agents (or organization) removes the limits of the cognitive capacities of these members by dividing the resolution of the problem according to the capacities of each one. An organization therefore has much more cognitive capacity than an isolated agent since it implements the capabilities of each agent of the organization to achieve these ends. [1]

In addition, agents not only use their own experiences to interact with the environment, they also use the experiences of other agents. [2]

1.2 Human behavior in emergency situation

During an emergency evacuation, there are three major factors to take into account: the type of danger that presents itself, the environment around us and the people who can help us (safety officer, safety officer evacuation . . .). But before establishing the different behaviors during an emergency, we must first know the direct reactions.

Indeed, when an alarm sounds, it must first correctly hear to react quickly to the situation but also understand it: we must be careful not to discredit an alarm because of "false alarms" past heard during exercises.

In the event of an alert, there are three behaviors that stand out: the waiting for instruction, evacuation of the building and the search for information to proportion its reaction according to various factors (location of the aggressor, the type of emergency , the risks. . .).

Agents' behaviours can be complex, for example, they will have preferences for choosing their exit paths in the event of a fire[4]:

Tableau 4 – Ordre des préférences pour la sélection des sorties

Préférences	La sortie est-elle visible ?	La sortie est-elle familière ?	Y a-t-il des conditions perturbatrices supplémentaires engendrées par le feu ?
1	Oui	Oui	Non
2	Non	Oui	Non
3	Oui	Non	Non
4	Oui	Oui	Oui
5	Non	Oui	Oui
6	Oui	Non	Oui
Pas de préférence	Non	Non	Non
	Non	Non	Oui

Figure 1.1: Order of preference for the selection of exits

It has been said above that emergency evacuation simulation exercises can be confusing when the alert is real because of the officer's reluctance to question whether the emergency is beautiful and well existing. But on the other hand these exercises are necessary because habits can take an important place during an emergency evacuation. Indeed, frightened people, to avoid further stress, prefer to take a route they know well and therefore a "usual route".

Emergency exits are therefore used less because officers have never used them before, but there are also other reasons different from habit or active use for not using an emergency exit[5]:

- Stressed agents can rest on the crowd and therefore follow it even if it is heading in the wrong direction.
- Emergency exits are not properly marked.
- We don't know what he is behind an emergency exit, which causes additional stress.
- Visible locks and appearance can cause a psychological barrier because they do not want to trigger a new alarm or because they are afraid of finding the door closed.

In addition, one example can be mentioned: the impossibility of opening the emergency exit of a gymnasium because of its poor position (on the stairs) has contributed to the compression of some people and resulted in their deaths (City College of New York). This is all the more dramatic when we know that the goal was to create a passage in case of an emergency situation in order to save lives[6].

To improve the evacuation time of officers in an establishment, a simple but equally effective idea is to leave the emergency doors open, in fact during a simulation in a school classroom, students were able to evacuate twice as quickly when the emergency door was open[7]. Moreover, this simulation shows us that the agents will give priority to the path with the most passage capacity and the fewest obstacles before taking into account the distance. Similarly, the width of the door is equally important for evacuation and to avoid clumping. Indeed, the wider the emergency exit, the more efficient and rapid the evacuation is[8].

After collecting enough information, the agent will choose the one he considers optimal in terms of distance, safety and emotional impact according to several factors related to the emergency, such as the type of attack[9]. There are three phases of reasoning in an emergency situation[10]:

- Avoidance phase: avoid the influx of people.
- Survival phase: stay hidden, communicate with law enforcement, put your phone in silent mode, be silent and think about the possibility of evacuating.
- Recovery phase: testify, provide psychological support and investigate.

To simulate the conditions during an emergency situation, there are different methods (life-size experiments, interviews with survivors, video surveillance cameras, etc.) but the virtual reality approach of putting an agent virtually in an emergency situation presents distorted results because of the "virtual" aspect of the simulation, which is similar to a video game[11].

In an emergency situation, it may be thought that "everyone for himself" dominates the agents, but on the contrary, selfish behaviours are rare and altruism is rather high according to the observations made on the videos[11].

1.3 Models governing agents in crisis situations

There are models in the literature that can help us model the movements of agents and crowds according to different factors. There are two different types of models, macroscopic models to simulate crowd movement using the characteristics of gas and liquid flows and microscopic models considering an individual agent such as a system particle[9].

The (microscopic) model of "Social Force" is the best known (DE-calibrated optimal parameters). The advantage of this model is that it reproduces the characteristics of a crowd while including evacuation speeds and a non-linear position distribution over time. But the pre-evacuation behaviour, which is also important to simulate the conditions as accurately as possible, is not included in this model[7].

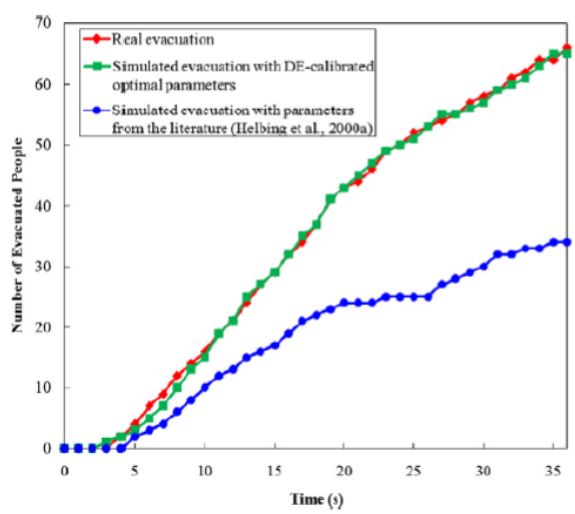


Figure 1.2: Number of evacuated people during the time

The Bayesian belief network (BBN) model is a method that takes into account uncertainties during decision-making, for example, the following diagram translates the BBN for the 2016 Paris attacks in a café.

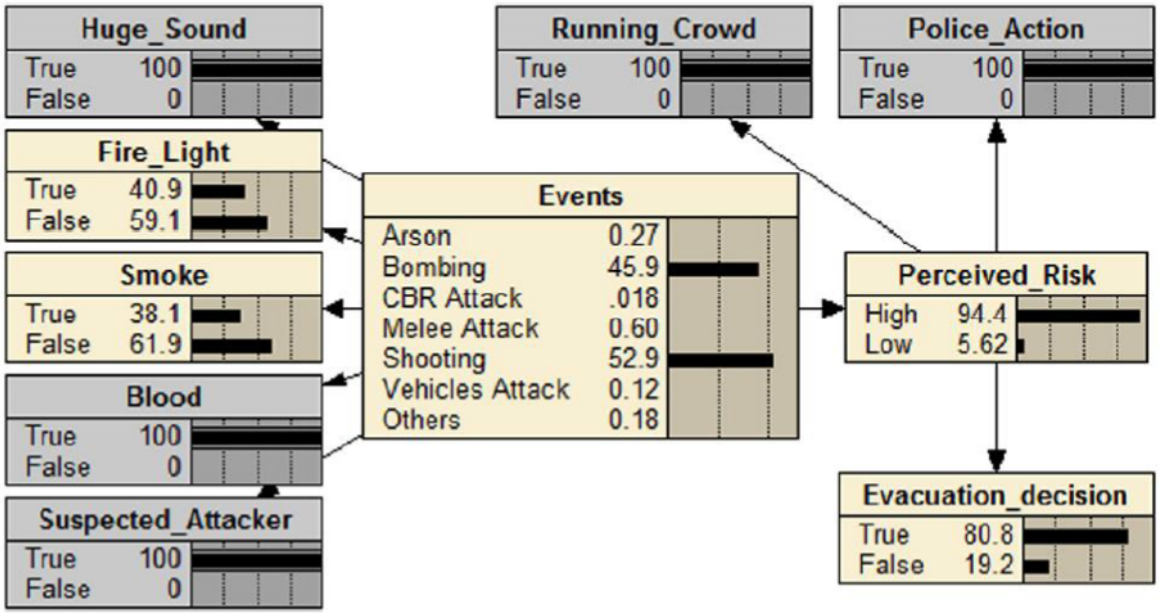


Figure 1.3: Bayesian belief network

This method takes into account the receipt of fuzzy information by agents and proceeds as follows in three stages: the assessment of the initial situation, the overall assessment phase and the choice of the path. During these steps, several algorithms are used but I will not detail them plus these methods[9].

Here is a diagram that summarizes the microscopic decision-making models that may exist:

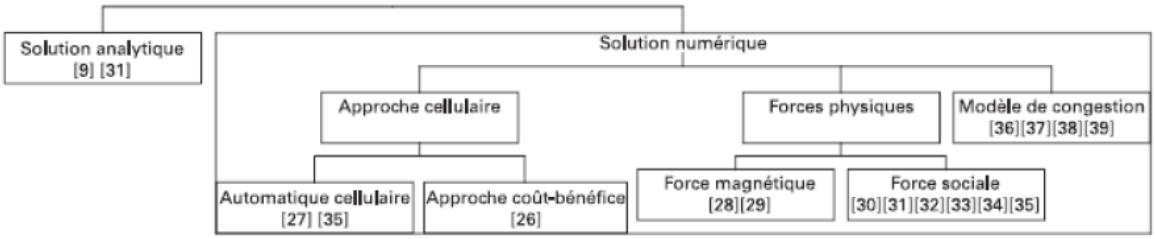


Figure 1.4: Microscopic decision-making models

There are also macroscopic models that can model a crowd in case of a situation as a matter of urgency. To do this, the literature compares the behaviour of a crowd to the characteristics of a gas: during an evacuation, individuals all project themselves towards the exit like atoms of gases that volatilize through a breach with more or less pressure (agglutination of individuals that cause a traffic jam). In addition, when an obstacle appears in the path of a crowd, the crowd separates on both sides and then joins each other once the obstacle has been overcome[12].

The ABM model (Braun, Bodmann and Musse[2005], Camillen et.al. [2009], Bonomi, Manzoni, Pisano and Vizzari[2009]) is used to study the evacuation of a crowd of establishment in various situations. This model has been used in particular to place the entrances/exits at optimal locations to facilitate evacuations from the shopping centre "COEX Mall" in South Korea (see map below)[13]:

Figure 1.5: COEX Mall

To link these two types of simulation (microscopic and macroscopic) we can decompose individuals according to their behaviours[4]:

- "Wait and see": Passive individuals who are waiting for leadership and who follow the crowd once the movement is created.
- "Information seeking": Individuals who try to understand the situation and seek information to modulate their reactions to the emergency.
- "Get out of here": Individuals whose first reaction is to flee the danger as quickly as possible.

In a crowd, the most important factors are: leadership, affiliation, supportive behaviours, communication and shared emotions. In addition, being part of a crowd can remove all the stress that can be associated with being alone in a dangerous situation.

Finally, the behaviour of each individual tells us whether he or she will rather follow the crowd or think individually even if the emotional factor may play a role in this decision making.

I consider that the models presented above are sufficient to provide an overview of the existing models even if the literature offers many others.

Chapter 2

Simulation of decision making

2.1 Graphical interface of the simulation

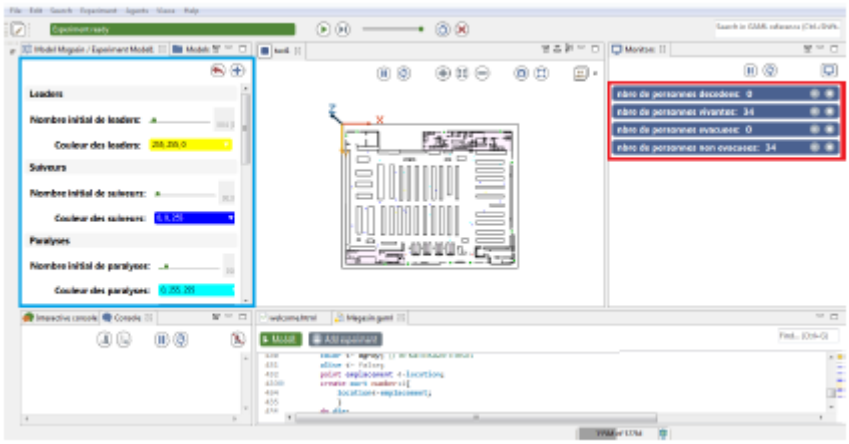


Figure 2.1: Graphical interface of the simulation

At the top is the start button, which will become a pause button if the simulation is started, a step-by-step button to perform the simulation step by step, a bar to select the minimum duration of a step, a button to reset the simulation and a button to close the simulation. The place where it says: "experiment ready" will display the number of steps performed during the simulation.

In the blue square on the left are the different variables of the simulation. They can take different values from one simulation to another.

In the centre is the map of the store, on which the agents move during the simulation. In the red box on the right are the real-time values of the following parameters: the number of people killed, the number of people alive, the number of people evacuated and the number of people not evacuated, i.e. still in the store. For each measure, terrorist agents and suicide bombers are not included.

At the bottom is the simulation code, written in GAML (GAma Modeling Language) as well as the heap size, it is a reference for the amount of memory used by the simulation objects. At the bottom left is the console.

2.2 Variables used in the simulation

There are five different classes of people: Leaders, Trackers, Paralyzed, Terrorists, Kamikazes

For these five classes, it is possible to select their colors as well as their number of individuals. It is also possible to set the speed and range of fire for terrorists as well as the speed and range of the suicide bomber's bomb. The leading, following and paralysed classes are made up of young, old and middle-aged people. For each of the first two categories, it is possible to indicate their percentages in the store population (the last percentage is deducted from the other two). It is also possible to indicate the speed of each category. I can also adjust their fields of vision.

Leaders are people who make decisions all the time, without following others. Trackers are those who follow others unless they see a closer exit.

Paralyses are like followers who paralyze themselves when they see a terrorist or suicide bomber.

I have set for individuals in the hiding places, in grey, one chance in two to hide. Moreover, once hidden, they have an 80% probability of remaining hidden at the next cycle. And finally, hidden individuals have a 50% chance of being seen by terrorists. These values can of course be modified in the code.

2.3 Other files used

.shp files are shapefiles, or "shape files". It is a file format for geographic information systems (GIS).

I used 4 of them, you can see them below:

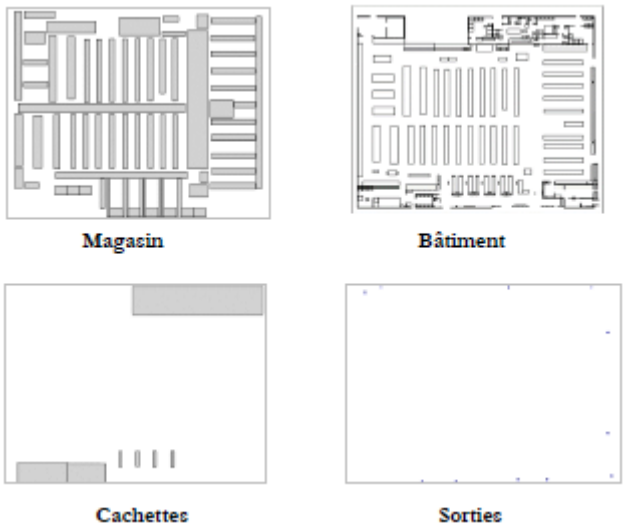


Figure 2.2: Shapefiles

The.shp store allows you to know the areas in which the leading, following and paralyzed agents can appear. Terrorists appear in the coordinate cell {450,450} and suicide bombers in the coordinate cell {220,450}.

The.shp building makes it possible to draw the boundaries of the building and consider them as obstacles. On this building, a 495*606 grid is applied. This corresponds to 1m on each side for each cell of the grid. The grid is not represented during the simulation in order to allow a more natural visualization of the movement.

The.shp cache allows you to trace the areas of the store in which the leading, following and paralyzed agents can hide. They are greyed out in the simulation (It is also possible to change these areas to a point that could attract individuals).

The.shp outputs are used to position the outputs of the store. They are shown in green in the simulation. They provide information on the number of people evacuated.

The simulation is divided into different layers. First of all, the walls of the store are marked out. They are considered as obstacles by the agents (building). Then, the locations in which the individuals appear (store) are added. Then the exits are plotted so that individuals can evacuate the store (exits). Finally, the hiding places are set up (hiding places). These four layers overlap during the simulation.

To create each layer, I used the computer-aided drawing software AutoCAD (Student Version) and an .dxf (Drawing eXchange Format) store plan.

For each layer, it was necessary to start from the original store drawing, then remove the unwanted parts and draw the desired points or areas.

Then, I transformed the files in.dxf format into.shp format files because GAMA only reads files in this format. For this I used CAD2Shape (Trial version) which allows us to convert according to the desired shape, for example for the file presenting the outputs, I needed points while for the cache file, I needed polygons. This is how I obtained the different layers while respecting the distances from the store.

2.4 Vision coding

In addition to these layers, the vision calculation of each individual agent is added. It is carried out during each movement of an individual agent, thus making it possible to regularly update its field of vision. To do this, a grid of a given size is drawn. It is centred on the agent. All the cells of the store in this grid are scanned. When an obstacle is detected, its box is considered invisible. It can hide other squares considered hitherto visible. The algorithm scans them again and makes them invisible.

Here is how our algorithm works for an individual (in blue) and an obstacle (in black) for each case (the grey boxes are not visible to the individual):

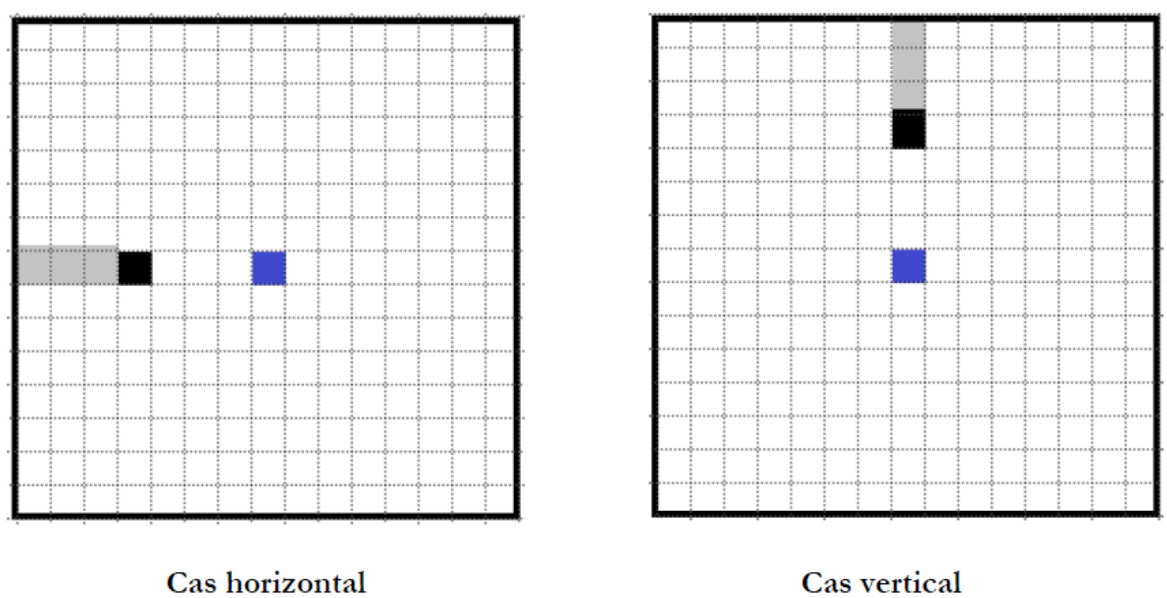
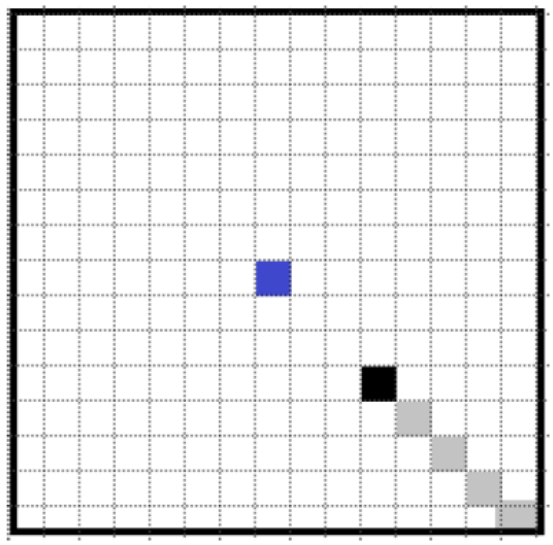


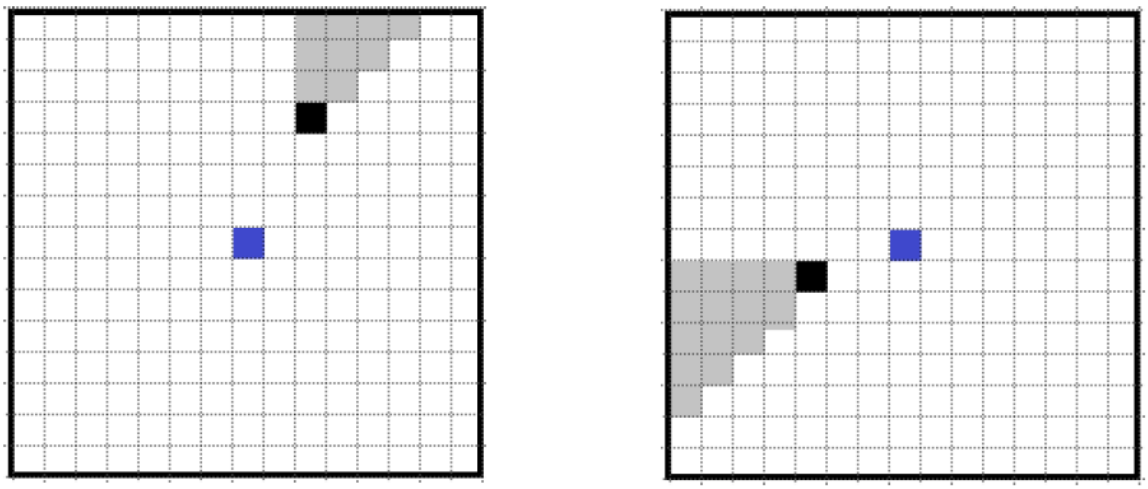
Figure 2.3: Horizontal and Vertical cases

In these examples, I took a field of view of 7 cells, which is equivalent to 7 meters in our simulation (I took 1 cell = 1 meter). As the human field of vision is much greater, it will have to be increased during simulations, which will lead to an increase in computing time. Indeed, the field of vision is recalculated for each individual, at each simulation cycle.



Cas diagonal

Figure 2.4: Diagonal case



Cas particuliers (2 exemples)

Figure 2.5: Random cases

Conclusion

The objective of simulating human behaviour during an emergency situation led us to use the GAMA development environment which allows us to process multi-agent systems. This environment allows us to observe the behaviour of the agents present in the store and more particularly the decisions they can make in these situations. The purpose of this simulation is to find more efficient evacuation solutions.

During this internship, I discovered multi-system agents and Gama platform with its language GAML.I also learnt some psychology concepts that are explained in the first part of this report. The code is not available because it is confidential. I also used the software AutoCAD in order to create an environment where the agents can evoluate during the simulation.

One of the most important improvement that could be done is to add machine learning. In that case, the agents could learn during the simulation and be more realistic.

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Abstract

During this intership, I developped my programming skill by working on a multi-agents system a by using the GAma Modeling Language. I also combined it with psychology concept in order to be more realistic. The code is not available because of confidentiality.

Keywords: GAML, Gama platform, psychology, multi-agents system, emergency, evacuation